

Cover Page

Course Title: Study Project

Project Title: Rydr: A Community-Driven Scenic Routing & Social Platform for Moto Riders

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1. Introduction

1.1 Purpose of Phase 2

The objective of Phase 2 is to translate the problem definition and requirements identified in Phase 1 into a concrete and well-defined system design. This phase focuses on defining the overall system architecture, functional and non-functional requirements in detail, and validating technical feasibility through a Proof of Concept (PoC). The phase ensures that the proposed solution for Rydr is practical, scalable, and ready for full implementation in Phase 3.

1.2 Scope of Phase 2

Phase 2 includes:

- Detailed system design and high-level architecture
 - Clearly defined functional and non-functional requirements
 - High-level database schema and data flow design
 - Development of a working Proof of Concept demonstrating core features
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2. System Overview

2.1 Product Perspective

Rydr is a standalone mobile application designed primarily for motorcycle riders, with future extensibility for other vehicle types. The system interacts with users through a mobile interface, backend services for data processing, and third-party services such as map and GPS APIs. The application is deployed as a mobile-first platform (Android), supported by cloud-based backend infrastructure.

2.2 Major System Functions

- User registration, authentication, and profile management

- Scenic route discovery and community-based route ratings
- Ride tracking and logging using GPS data
- Social features including posts, likes, comments, and discussions
- Milestones, badges, and leaderboards for gamification

2.3 User Classes and Characteristics

- **Riders (End Users):** Motorcycle riders who discover routes, log rides, and engage with the community
 - **Administrators:** Manage content moderation, reports, and system maintenance
 - **External Services:** Map, GPS, and media storage APIs integrated into the system
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3. Functional Requirements

- **FR1:** The system shall allow users to register, log in, and manage rider profiles with bike details.
 - **FR2:** The system shall recommend scenic routes based on community data and tags.
 - **FR3:** The system shall allow users to log rides and track distance, duration, and elevation.
 - **FR4:** The system shall support photo, video, and story sharing related to rides.
 - **FR5:** The system shall enable discussion threads, likes, and comments for community interaction.
 - **FR6:** The system shall provide leaderboards, milestones, and badges for user engagement.
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4. Non-Functional Requirements

4.1 Performance Requirements

- The system should respond to user actions within 2 seconds under normal load.
- Real-time GPS tracking should update location data with minimal delay.

4.2 Security Requirements

- Secure authentication using token-based mechanisms.
- Protection of user data through encryption and access control.

4.3 Usability Requirements

- Rider-friendly UI with minimal interactions and clear navigation.
- Accessibility considerations for outdoor and riding conditions.

4.4 Scalability and Maintainability

- Modular architecture supporting future feature extensions.
 - Cloud-based backend allowing gradual regional to global scaling.
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5. System Architecture and Design

5.1 System Architecture Diagram

The system follows a client-server architecture consisting of a mobile client, backend services, databases, and third-party APIs. The mobile app communicates with backend APIs for authentication, route management, and social features, while map and GPS services provide location data.

5.2 Module-wise Design

- **User Module:** Handles authentication, profiles, and user data.
- **Route Module:** Manages scenic routes, tags, ratings, and recommendations.
- **Ride Tracking Module:** Captures GPS data and computes ride metrics.
- **Social Module:** Supports posts, comments, likes, and discussions.
- **Gamification Module:** Manages milestones, badges, and leaderboards.

5.3 Data Flow Design

User inputs from the mobile app are sent to backend services for processing and storage. GPS and map data are fetched from external APIs, processed for analytics, and stored in the database for retrieval and visualization.

5.4 Database Design

Key entities include Users, Routes, Rides, Posts, Comments, and Achievements. Relationships connect users to rides, routes, and social interactions.

6. Technology Stack and Justification

Frontend: Android (React Native) for cross-platform development.

Backend: FastAPI, REST/GraphQL

Database: Postgres SQL

Other Tools: CDN, MapBox, etc.

7. Proof of Concept (PoC)

7.1 PoC Description

The PoC implements core features including user authentication, basic rider profiles, route display on maps, and ride logging. It validates technical feasibility and integration of GPS and map services.

7.2 PoC Demonstration Details

- Demonstration via screenshots or emulator runs
 - Features shown: login, route view, ride tracking
 - Limitations: limited route intelligence and basic UI
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8. Testing and Validation Strategy

- Unit testing for core modules
 - Integration testing for API and GPS services
 - Manual testing on Android devices and emulators
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9. Risks, Challenges, and Mitigation

Risks:

- GPS inaccuracies
- Dependency on third-party APIs
- Limited development time

Mitigation:

- Use averaged GPS sampling
 - Modular API integration
 - Focused MVP-based development
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10. Phase 2 Outcomes and Readiness for Phase 3

Phase 2 delivers a complete system design, validated requirements, and a working PoC. The project is ready to move into Phase 3 for full-scale implementation and feature completion.

11. Supervisor Review and Approval

Advisor Feedback:

Supervisor Comments:

Recommendations:

Signature: _____

Date: _____